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ECON 308 – International Economic Relations
MIDTERM EXAM #2
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Instructions:

1. You have 75 minutes to work on the exam.
2. This midterm is individual and closed-book.
3. For the multiple-choice questions, choose the option that best answers/completes the question.
4. This Examination paper includes 8 pages (includes the cover page). You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attentions of your invigilator.

PLEASE: PUT YOUR NAME AND STUDENT ID ON EVERY SHEET!!

Multiple Choice Answers:

- | | |
|-----------|-----------|
| 1. _____ | 15. _____ |
| 2. _____ | 16. _____ |
| 3. _____ | 17. _____ |
| 4. _____ | 18. _____ |
| 5. _____ | 19. _____ |
| 6. _____ | 20. _____ |
| 7. _____ | 21. _____ |
| 8. _____ | 22. _____ |
| 9. _____ | 23. _____ |
| 10. _____ | 24. _____ |
| 11. _____ | |
| 12. _____ | |
| 13. _____ | |
| 14. _____ | |

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question (1 mark each, total of 24 marks).

- 1) Trade benefits a country by
- A) increasing the wage rate.
 - B) increasing available consumption choices.
 - C) increasing the real income of all resource owners.
 - D) reducing the relative price of the exported good.
 - E) reducing the need for specialization in production.
- 2) In the U.S. economy over the last half-century,
- A) trade-based technological change has been associated with a lower relative cost of skilled workers, leading to increasing skilled-unskilled employment ratios in all sectors.
 - B) trade-based technological change has been associated with a higher relative price of high-tech goods, leading to increasing skilled-unskilled employment ratios in all sectors.
 - C) skilled-based technological change has been associated with a higher skilled-unskilled wage ratio, leading to increasing skilled-unskilled employment ratios in all sectors.
 - D) skilled-based technological change has been associated with a higher skilled-unskilled wage ratio, leading to decreasing skilled-unskilled employment ratios in all sectors.
- 3) International trade has strong effects on income distributions. Therefore, international trade
- A) will tend to hurt some groups in each trading country.
 - B) will be beneficial to all those engaged in international trade.
 - C) will tend to hurt everyone in both countries.
 - D) is beneficial to everyone in both trading countries.
 - E) will tend to hurt one trading country.
- 4) Empirical observations on actual North-South trade patterns tend to
- A) support the validity of the Rybczynski Theorem.
 - B) support the validity of the Leontief Paradox.
 - C) support the validity of the wage equalization theorem.
 - D) support the validity of the neo-imperialism exploitation theory.
 - E) support the validity of the Heckscher-Ohlin model.

- 5) An economy with a high relative supply of labor to capital will be relatively better at
- A) producing food than an economy with a low relative supply of labor to capital.
 - B) producing food than an economy with a high relative supply of capital to labor.
 - C) producing both cloth and food than an economy with a low relative supply of labor to capital.
 - D) producing cloth than an economy with a low relative supply of labor to capital.
- 6) In the 2-factor, 2-good Heckscher-Ohlin model, the country with a relative abundance of _____ will have a production possibility frontier that is biased toward production of the _____ good.
- A) land; capital intensive
 - B) capital; capital intensive
 - C) land; labor intensive
 - D) labor; land intensive
 - E) labor; capital intensive
- 7) In the 2-factor, 2-good Heckscher-Ohlin model, an influx of workers from across the border would
- A) shift the production possibility curve outward and decrease the production of the labor-intensive product.
 - B) shift the production possibility curve outward, and increase the production of both goods.
 - C) move the point of production along the production possibility curve.
 - D) shift the possibility curve outward and displace preexisting labor.
 - E) shift the production possibility curve outward and much larger in the direction of the labor-intensive product than in the direction of the capital-intensive product.
- 8) In the Heckscher-Ohlin model, comparative advantage is influenced by the interaction between nations' resources, called _____, and the technology of production, called _____, used in the production of different goods.
- A) factor resources; factor technology
 - B) interactive resource; interactive technology
 - C) relative abundance; relative intensity
 - D) factor abundance; factor intensity
- 9) The Heckscher-Ohlin theory is also referred to as the
- A) endowment theory.
 - B) relative abundance theory.
 - C) resource theory.
 - D) factor-proportions theory.
- 10) The direct effect of a tariff is to make
- A) imported goods cheaper inside a country than they are outside the country.
 - B) imported goods more expensive inside a country than they are outside the country.
 - C) imported goods the same price inside a country as they are outside the country.
 - D) imported goods more expensive outside a country than they are inside the country.

11) The distinctive feature of tariffs and export subsidies is that

- A) they give incentives for importers and exporters.
- B) they create a difference between prices at which goods are traded on the world market and prices at which those goods can be purchased within a country.
- C) they equalize prices at which goods are traded on the world market and prices at which those goods can be purchased within a country.
- D) they make imported and exported goods cheaper.

12) A country will be able to consume a combination of goods that is NOT attainable solely from domestic production if

- A) the world terms of trade equal the domestic relative costs.
- B) the country avoids international trade.
- C) the country's domestic production value equals world relative value.
- D) the country specializes in one product.
- E) the world terms of trade differ from its domestic relative costs.

13) Which of the following statements is NOT true?

- A) Export-biased growth in the U.S. improves the U.S.'s terms of trade.
- B) Export-biased growth in the U.S. worsens the U.S.'s terms of trade.
- C) Export-biased growth in the rest of the world improves the U.S.'s terms of trade.
- D) Import-biased growth in the U.S. improves the U.S.'s terms of trade.
- E) Import-biased growth in the rest of the world worsens the U.S.'s terms of trade.

14) A geographically concentrated industry can facilitate _____ that allow informal exchange of information and ideas at a personal level.

- A) knowledge spillovers B) technical gossiping
- C) creative commons D) social gathering

15) In the presence of external economies of scale, trade

- A) will unambiguously worsen welfare in the exporting country and improve welfare in the importing country.
- B) will unambiguously improve welfare in the exporting country and worsen welfare in the importing country.
- C) will unambiguously improve welfare in both countries.
- D) may or may not improve welfare in both countries.
- E) will unambiguously worsen welfare in both countries.

16) If a firm's output less than doubles when all inputs are doubled, production is said to occur under conditions of

- A) imperfect competition.
- B) increasing returns to scale.
- C) decreasing returns to scale.
- D) constant returns to scale.
- E) intra-industry equilibrium.

17) Internal economies of scale will _____ cost per unit when output is _____ by _____.

- A) reduce; reduce; the industry
- B) reduce; increased; a firm
- C) reduce; increased; the industry
- D) increase; increased; a firm
- E) increase; increased; the industry

18) In presence of external economies of scale, a country's industry with _____ cumulative output to date will experience a _____ unit cost.

- A) lower; lower
- B) higher; higher
- C) higher; lower
- D) higher; constant

19) When economies of scale apply at the level of the industry rather than at the level of the individual firms, they are called

- A) mixed economies.
- B) scope economies.
- C) internal economies.
- D) external economies.
- E) scale economies.

20) Apart from comparative advantage, _____ can play a key role in determines the pattern of specialization and trade in industries with external economies of scale.

- A) historical accident
- B) natural disasters
- C) civil wars
- D) decreasing returns to scale

21) External economies of scale arise when the cost per unit

- A) falls as the industry and the average firm grows larger.
- B) remains constant over a broad range of output.
- C) rises as the industry grows larger and falls as the average firm grows larger.
- D) falls as the industry grows larger and rises as the average firm grows larger.
- E) rises as the industry and the average firm grows larger.

22) One advantage of the specialization that results from international trade is that countries can take advantage of

- A) taste reversals.
- B) production diversification.
- C) scale economies.
- D) smaller countries.
- E) lower transport costs.

23) Because of forward-falling supply curve, China's _____ production output as a result of trade leads to button price that is _____ than the price before trade.

- A) lower; lower
- B) higher; lower
- C) lower; higher
- D) higher; higher

24) A cluster of firms may be more efficient than an individual firm in isolation because it can take advantage of

- A) specialized suppliers, government policies, geographical location.
- B) labor market pooling, tax policies, internal economies of scale.
- C) knowledge spillovers, network externalities, government support.
- D) specialized suppliers, labor market pooling, and knowledge spillovers.

Problem 1 (13 marks)

Two countries, **Home (H)** and **Foreign (F)**, produce two goods: **textiles (T)** (labour-intensive) and **steel (S)** (capital-intensive). Both countries have the same technologies and homothetic preferences.

Country	Capital K	Labour L	K/L ratio
Home	300	900	0.33
Foreign	800	800	1.00

Assume perfect competition, identical production functions, and full employment.

- (1) Identify which country is relatively labour-abundant and which is capital-abundant.
- (2) According to the Heckscher–Ohlin theorem, which good will each country export under free trade?
- (3) Using the Stolper–Samuelson theorem, explain how opening to trade affects the real wage and the real rental rate in each country.
- (4) Suppose the world relative price of textiles (P_T/P_S) rises by 10%. Explain how this change affects factor incomes (wages and rents) and the output composition in the Home country.
- (5) If trade continues over time, what mechanism could lead to **factor price equalization** between the two countries?

Problem 2 (13 marks)

A country produces **agricultural products (A)** and **electronics (E)** with the production possibility frontier (PPF):

$$A^2 + 2E^2 = 900.$$

The world relative price is $P_E/P_A = 3$. The country consumes at the point where preferences satisfy $E = 0.5A$.

- (1) Determine the production point (A_P, E_P) that maximizes the value of output at world prices.
- (2) Find the consumption bundle (A_C, E_C) , identify which good is exported, and compute the value of exports and imports.
- (3) Verify that trade is balanced at world prices.
- (4) Explain in words how an increase in P_E/P_A (improvement in the terms of trade) affects the country's production, consumption, and welfare.
- (5) Suppose the economy experiences **export-biased growth** that expands production possibilities more in electronics than in agriculture.
 - Describe how this changes the country's trade pattern and the direction of the terms-of-trade effect.
 - Discuss whether welfare necessarily increases after such growth.